The Department of Computer Applications

Course Level Problem Based Learning

Course Title: Cyber Security

Course Code: 24MC205B

Semester: **Second Semester**

A CLPBL Project Report Titled

SafeConnect – AI Powered Child Safety Monitoring System

Submitted By

Sl No.	USN	Name
1	4SO25MC071	Poorvi Bhandary
2	4SO25MC101	Supritha N K
3	4SO25MC105	Vaishali B

Under the Guidance of

Akshatha N K

Assistant Professor

10 July 2026

ST JOSEPH ENGINEERING COLLEGE

MANGALURU – 575 028

The Department of Computer Applications

CERTIFICATE OF COMPLETION

It is hereby certified that this CLPBL Project Report is a bonafide work carried out by the students listed below under my guidance in the Department of Computer Applications. This work has been completed towards the partial fulfilment of the course Cyber Security with course code 24MC205B, offered in the Second Semester.

It is further certified that the students have followed the Design Thinking framework throughout the execution of the project and have incorporated all the corrections/suggestions provided during the progress reviews and assessments. The project work has been evaluated and approved as it meets the requirements of CLPBL Implementation as mandated by the Department.

Sl No.	USN	Name
1	4SO25MC071	Poorvi Bhandary
2	4SO25MC101	Supritha N K
3	4SO25MC105	Vaishali B

Ms. Akshatha N K
Assistant Professor
Department of Computer Applications

Mr. Harish B
Head of the Department
Department of Computer Applications

ST JOSEPH ENGINEERING COLLEGE

MANGALURU – 575 028

The Department of Computer Applications

DECLARATION

We, the undersigned students of the Department of **Computer Applications**, hereby declare that the project work entitled **"SafeConnect –AI Powered Child Safety Monitoring System"**, submitted in partial fulfilment of the requirements of the course **Cyber Security** with course code **24MC205B**, in the **Second Semester**, is an original **Course Level Problem Based Learning (CLPBL)** project carried out by us.

We confirm that this project has been completed using the **Design Thinking framework** under the guidance of **Ms Akshatha N K**. We also certify that the work reported in this document has not been previously submitted to any institution or organization for any academic award, degree, or certification. We abide by the academic integrity and ethical standards of the Department and the Institution, and accept that any violation, if discovered later, may lead to appropriate action.

Students' Signatures:

Sl No.	USN	Name	Signature
1	4SO25MC071	Poorvi Bhandary	
2	4SO25MC101	Supritha N K	
3	4SO25MC105	Vaishali B	

Date: 10 July 2026

Place: Mangaluru

ACKNOWLEDGEMENT

We express our sincere gratitude to **Ms. Akshatha N K**, Guide for the Course Level Problem Based Learning (CLPBL) project, Department of Computer Applications, St Joseph Engineering College, Mangaluru, for her valuable guidance, continuous support, and encouragement throughout the development of our project titled **"SafeConnect – AI Powered Child Safety Monitoring System."**

We extend our heartfelt thanks to **Mr. Harish B**, Head of the Department, Department of Computer Applications, for providing the necessary facilities, motivation, and support required for the successful completion of this project.

We also express our gratitude to all the faculty members of the Department of Computer Applications for their suggestions, constructive feedback, and encouragement during various stages of the project.

We would like to thank the management and administration of St Joseph Engineering College for providing an excellent learning environment and the resources needed to carry out this work.

Finally, we express our sincere appreciation to our parents, friends, and classmates for their constant support, encouragement, and cooperation throughout the project.

Team Members:

USN	Name
4SO25MC071	Poorvi Bhandary
4SO25MC101	Supritha N K
4SO25MC105	Vaishali B

ABSTRACT

SafeConnect is an AI-powered child safety monitoring system developed to address the growing concerns surrounding cyberbullying, online harassment, and suspicious online interactions involving children. The project aims to provide a secure communication platform where children's conversations can be monitored for potentially harmful content while allowing parents to receive alerts regarding suspicious activities. The system was designed and implemented using Python, Streamlit, and JSON-based data storage. Separate login portals were developed for children and parents, enabling secure access to their respective dashboards. The child dashboard provides a simple chat interface, while the parent dashboard displays alerts, risk indicators, and monitored conversations. The project followed the Design Thinking framework to understand user needs, define the problem, develop suitable solutions, and test the prototype. The implemented solution demonstrates how intelligent monitoring syst ems can contribute to safer online environments for children. Future enhancements include integration of advanced Artificial Intelligence models such as Toxic-BERT for real-time toxicity detection, cloud-based storage, and mobile application support.

Keywords: Child Safety, Cyber Security, Cyberbullying Detection, Parent Monitoring, Streamlit, Artificial Intelligence

Table of Contents

Chapter – 1: Problem Description

With the rapid growth of digital communication, children are increasingly using social media platforms, messaging applications, and online gaming services for learning and interaction. While these platforms offer many benefits, they also expose children to cyberbullying, online harassment, inappropriate content, and interactions with unknown individuals. Such online threats can negatively affect a child's mental health, confidence, academic performance, and overall well-being.

Many parents are concerned about their children's online safety but find it difficult to continuously monitor their digital activities due to time constraints, privacy concerns, and the large volume of online communication. Existing parental control applications mainly focus on screen-time management, website blocking, and location tracking, but they often fail to identify harmful or suspicious conversations in real time.

The problem addressed in this project is the lack of an intelligent and user-friendly system that can automatically monitor online conversations, detect potentially harmful messages, and notify parents whenever suspicious activities are identified. To address this issue, the proposed **SafeConnect – AI Powered Child Safety Monitoring System** provides a secure communication platform with AI-based monitoring and alert generation, helping parents protect their children while promoting safer online interactions.

Chapter – 2: Key Findings of Secondary Research

A review of research papers, journals, articles, and online resources revealed that cyberbullying and online harassment are among the most significant challenges affecting children and teenagers in today's digital world. Studies indicate that excessive exposure to harmful online interactions can lead to anxiety, stress, depression, reduced self-confidence, and poor academic performance.

Existing parental control applications mainly provide features such as screen-time management, website filtering, location tracking, and application blocking. However, many of these systems lack the ability to intelligently analyze conversations and detect harmful or abusive messages in real time.

Research in the fields of Artificial Intelligence (AI) and Natural Language Processing (NLP) shows that machine learning models can effectively identify toxic language, cyberbullying, hate speech, and suspicious communication patterns. AI-based monitoring systems can provide early warnings to parents and help prevent serious online threats before they escalate.

The findings from the secondary research highlighted the need for an intelligent child safety monitoring system that combines secure communication with automated threat detection and parental alerts. These observations formed the foundation for developing the proposed **SafeConnect – AI Powered Child Safety Monitoring System**.

Chapter – 3: Key Findings of Primary Research

A majority of parents expressed concern about their children's exposure to cyberbullying, online harassment, and interactions with strangers on digital platforms.

- Most parents found it difficult to continuously monitor their children's online activities due to busy schedules and privacy concerns.
- Students reported that they frequently use messaging applications and social media for communication, making online safety an important issue.
- Teachers emphasized the need for awareness and early detection of harmful online behaviour to protect children from cyber threats.
- Parents preferred a system that could automatically detect suspicious messages and notify them instead of manually checking chat histories.
- Users suggested that the application should have a simple, secure, and user-friendly interface for both parents and children.
- Respondents appreciated the idea of receiving timely alerts for potentially harmful conversations while maintaining children's privacy.
- The feedback collected confirmed the need for an AI-powered child safety monitoring system that provides real-time monitoring and early threat detection.

Chapter – 4: Problem Definition

Children are increasingly exposed to cyberbullying, online harassment, inappropriate content, and suspicious interactions through digital communication platforms. These online threats can have a serious impact on their mental health, emotional well-being, and overall development. Although parents are concerned about their children's online safety, continuously monitoring online conversations is difficult due to time constraints, privacy concerns, and the large volume of digital communication. Most existing parental control applications focus on restricting access to websites or managing screen time, but they do not effectively detect harmful conversations or provide timely alerts.

Therefore, there is a need for an intelligent, secure, and user-friendly child safety monitoring system that can automatically analyze chat messages, identify suspicious or abusive content, and notify parents whenever potential threats are detected. The proposed **SafeConnect – AI Powered Child Safety Monitoring System** addresses this need by combining AI-based message monitoring, threat detection, and real-time parental alerts to create a safer online environment for children.

Chapter – 5: Project Objectives

The primary objectives of the **SafeConnect – AI Powered Child Safety Monitoring System** are:

- To develop a secure and user-friendly communication platform for children and parents.
- To monitor chat messages and identify suspicious, abusive, or harmful content using Artificial Intelligence.
- To provide timely alerts to parents whenever potential online threats are detected.
- To maintain a secure chat history for monitoring and future reference.
- To enhance children's online safety by enabling early detection of cyberbullying and inappropriate interactions.
- To create separate dashboards for children and parents with secure login authentication.
- To provide a simple, efficient, and reliable solution that promotes safer online communication while respecting user privacy.

Chapter – 6: Business Model Canvas

Value Proposition:

- AI-powered monitoring of child chats:
    - Uses Artificial Intelligence to analyze children's chat messages and identify suspi-cious, harmful, or inappropriate conversations.
- Real-time cyberbullying detection:
    - Detects abusive, offensive, or bullying messages as they occur and helps prevent online harassment.
- Online grooming detection:
    - Identifies suspicious conversations that may indicate attempts by strangers to manipu-late or exploit children.
- Toxic language detection using Toxic-BERT:
    - Uses the Toxic-BERT model to recognize toxic, offensive, and abusive language in chat messages.
- Instant parent alerts with recommendations:
    - Sends immediate notifications to parents when harmful messages are detected and provides suggested actions.
- Privacy-focused monitoring (only flagged messages shown):
    - Displays only suspicious messages to parents, helping protect the child's privacy while ensuring safety.

Customer Segments:

- Parents:
  Parents use the system to monitor their children's online activities and receive alerts about potential cyber threats.
- Schools:
  Schools can use the system to promote a safe digital environment and help protect students from cyberbullying.
- Educational Institutions:
  Educational institutions can adopt the system to enhance cybersecurity awareness and support student online safety initiatives.
- Child Safety Organizations:
  Child safety organizations can use the platform to monitor online risks, support child protection efforts, and create awareness about digital safety.

Key Activities:

- Secure User Authentication:
  Provides a secure login system to ensure that only authorized parents and children can access the application.
- Chat Monitoring:
  Continuously monitors chat messages to identify harmful or suspicious conversations.
- AI-based Message Classification:
  Uses the Toxic-BERT AI model to classify messages as normal or potentially harmful based on their content.
- Rule-based Threat Detection:
  Applies predefined rules to detect threats such as cyberbullying, toxic language, and online grooming attempts.
- Alert Generation:
  Generates instant alerts and notifications for parents whenever suspicious or harmful messages are detected.
- Parent Dashboard Reporting:
  Displays flagged messages, alerts, and monitoring reports through a dedicated parent dashboard for easy review.

Key Resources:

- Python:
  Python is the primary programming language used to develop the application's backend logic and AI functionalities.
- Streamlit:
  Streamlit is used to build the web-based user interface, allowing parents and children to interact with the system easily.
- Hugging Face Toxic-BERT:
  The Toxic-BERT model is used to analyze chat messages and detect toxic, abusive, or harmful language using Artificial Intelligence.
- JSON Database:
  JSON is used as a lightweight database to store user information, chat messages, and alert data.
- Visual Studio Code:
  Visual Studio Code is the integrated development environment (IDE) used for writing, testing, and debugging the project code.

Customer Relationships:

- Continuous Monitoring:
  The system continuously monitors chat messages to detect harmful or suspicious activities and ensure child safety.
- AI-assisted Safety:
  Artificial Intelligence helps identify cyberbullying, toxic language, and other online threats with improved accuracy.
- Real-time Alerts:
  The system immediately notifies parents whenever suspicious or harmful messages are detected.
- Secure Parent Access:
  Parents can securely log in to view alerts, flagged messages, and monitoring reports while protecting user privacy.

Channels:

- Web Application:
  The SafeConnect system is provided as a web application that allows parents and children to securely access the platform through a web browser.
- Educational Institutions:
  Schools and colleges can adopt and promote the system to enhance students' online safety and digital well-being.
- Parent Awareness Programs**:**
  The system can be introduced through awareness programs, workshops, and seminars to educate parents about online child safety and the benefits of AI-based monitoring.

Cost Structure:

- Development:
  The cost involved in designing, coding, testing, and developing the SafeConnect application.
- AI Model Execution:
  The resources required to run the Toxic-BERT model for analyzing chat messages and detecting harmful content.
- Internet:
  An internet connection is required for accessing the web application and supporting AI-based message processing.
- Maintenance:
  The cost of updating the system, fixing bugs, improving features, and ensuring smooth performance after deployment.

Revenue Streams (Future Possibilities):

- Premium Subscriptions:
  Parents can subscribe to premium plans to access advanced monitoring features, detailed reports, and additional security options.
- School Licenses:
  Schools can purchase licenses to use the SafeConnect system for monitoring and protecting students in their institution.

- Enterprise Monitoring Solutions:
  Organizations and child safety agencies can use customized enterprise solutions for large-scale child safety monitoring and reporting.
- Cloud Hosting Services:
  The system can be offered as a cloud-based service, where users pay for secure online access, data storage, and continuous system availability.

Chapter –7. SWOT Analysis

Strengths:

- AI-powered Threat Detection:
  Uses Artificial Intelligence to identify harmful and suspicious messages with greater accuracy.
- Separate Child and Parent Dashboards:
  Provides dedicated dashboards for children and parents, ensuring secure and role-based access.
- Real-time Monitoring:
  Monitors chat messages continuously and detects threats as they occur.
- Toxic-BERT Integration:
  Uses the Toxic-BERT model to accurately detect toxic, offensive, and abusive language.
- Online Grooming Detection:
  Identifies suspicious conversations that may indicate attempts to manipulate or exploit children.

Weaknesses:

- Uses JSON instead of SQL Database:
  The system stores data in a JSON file, which is suitable for small projects but less efficient and scalable than an SQL database.
- Prototype with Simulated Chats:
  The application has been tested mainly with simulated chat conversations rather than real-world user data.

- English Language Support Only:
  The current version analyzes messages only in English and does not support multiple languages.
- Limited AI Training Dataset:
  The AI model is trained on a limited dataset, which may reduce its accuracy when detecting new or complex online threats.

Opportunities:

- Mobile App:
  The system can be expanded into a mobile application for easier access and monitoring on smartphones.
- Cloud Database:
  Using a cloud database can improve data storage, scalability, and secure access from multiple devices.
- Live Notifications:
  The system can send instant notifications to parents whenever harmful or suspicious messages are detected.
- School Deployment:
  The application can be implemented in schools to improve students' online safety and digital well-being.

Threats:

- Privacy Concerns:
  Users may have concerns about the security and privacy of children's personal and chat data.
- Evolving Cyber Threats:
  New cyber threats and attack techniques continue to emerge, requiring the system to be updated regularly.
- False Positives:
  The AI model may sometimes classify harmless messages as harmful, leading to unnecessary alerts.
- Internet Dependency:
  The system requires a stable internet connection for smooth operation and real-time monitoring.

Chapter – 8: Description of Work

- The SafeConnect system was developed following the Design Thinking methodology consisting of Empathize, Define, Ideate, Prototype, and Test phases.
- The project aims to improve child online safety by continuously monitoring chat conversations and identifying suspicious messages using Artificial Intelligence and rule-based Natural Language Processing techniques.
- The application consists of two independent dashboards:
  - Child Dashboard
  - Parent Dashboard
- The child can send and receive messages through a secure chat interface, while the parent dashboard automatically analyzes every conversation and displays alerts whenever harmful communication is detected.
- The backend continuously stores chat history in JSON format and performs AI-based threat analysis before presenting results to parents

8.1 Ideation and Design

Several approaches were evaluated including:

- Manual parental monitoring
- Browser extensions
- Keyword filtering
- AI-powered message monitoring
- The AI-powered solution was selected because it provides intelligent detection instead of relying solely on manual supervision.

- The final system architecture consists of:

  1. Login Module:

  Provides secure authentication for both parents and children, allowing only authorized users to access the system:

  2. Child Dashboard:

  Offers a simple and user-friendly interface where children can securely send and receive chat messages:

3. **Parent Dashboard:**

Allows parents to view alerts, flagged messages, threat reports, and monitor their child's online safety:

4. **Chat Storage Module:**

Stores chat messages and user information securely in a JSON database for analysis and future reference.

5. **AI Detection Engine:**

Uses the Toxic-BERT model to analyze chat messages and detect toxic, abusive, or harmful content using Artificial Intelligence.

6. **Rule-Based Detection Engine:**

Applies predefined rules and keywords to identify cyberbullying, online grooming, and other suspicious activities.

7. **Alert Generation Module:**

Automatically generates alerts and notifies parents whenever harmful or suspicious messages are detected.

8. **Threat Classification Module:**

Classifies detected messages into different threat categories and determines their severity to help parents understand the level of risk.

8.2 Prototyping / Simulation

The prototype was developed using:

Software Used:

- Python:

  Python is the main programming language used to develop the application and implement the AI-based functionalities.
- Streamlit:

  Streamlit is used to create the interactive web interface for the child and parent dashboards.

- Visual Studio Code:
  Visual Studio Code is the development environment used for writing, testing, and debugging the project code.
- Hugging Face Transformers:
  The Hugging Face Transformers library is used to load and run the Toxic-BERT model for detecting toxic and harmful messages.
- JSON Database:
  JSON is used as a lightweight database to store user details, chat messages, and alert information.

AI Models Used:

- **Toxic-BERT (unitary/toxic-bert):**
  Toxic-BERT is a pre-trained AI model used to identify toxic, abusive, offensive, and harmful language in chat messages.

Modules Implemented:

- Parent Login:
  Allows parents to securely log in and access alerts, reports, and monitoring features.
- Child Login:
  Provides secure authentication for children to access the chat application.
- Child Chat Interface:
  Offers a simple and user-friendly chat interface where children can send and receive messages.
- Message Storage:
  Stores chat messages and user information securely in a JSON database for future analysis.
- AI Detection:
  Uses the Toxic-BERT model to analyze chat messages and identify harmful or toxic content.
- Cyberbullying Detection:
  Detects bullying, harassment, and abusive messages to help protect children from online abuse.

- Online Grooming Detection:
  Identifies suspicious conversations that may indicate attempts to manipulate or exploit children online.
- Toxic Language Detection:
  Recognizes offensive, hateful, or inappropriate language using Artificial Intelligence.
- Parent Alert Dashboard:
  Shows flagged messages and real-time alerts, enabling parents to quickly respond to potential risks.

Working of the AI Detection Module:

Whenever a message is sent:

- The message is stored in messages.json.
- The system immediately sends the message to the AI Detection Module.

The Rule-Based Engine first checks for predefined cyberbullying and online grooming phrases.

Examples:

- Cyberbullying:
  Go away
  Nobody wants you
  You're disgusting
  Dumb
- Online Grooming:
  Send your photo
  Meet me
  Don't tell your parents
  Where do you live

If no rule matches, the message is analyzed using the Toxic-BERT model.

Toxic-BERT predicts whether the message is toxic along with a confidence score.

If the confidence exceeds the threshold, the message is marked as suspicious.

The Parent Dashboard automatically displays:

- Threat Type
- Sender
- Timestamp

- AI Confidence
- Severity
- Recommended Action

8.3 Testing

The system was tested using multiple chat scenarios.

- Test Case 1

  Message:

  Hello, how are you?

  Expected Result:

  Safe

  Actual Result:

  No alert generated

- Test Case 2

  Message:

  Go away, nobody wants you here.

  Expected Result:

  Cyberbullying detected

  Actual Result:

  Alert generated

  Severity:

  Critical

- Test Case 3

  Message:

  Send your photo.

  Expected Result:

  Online Grooming

  Actual Result:

  Alert generated

- Test Case 4

Message:

You are disgusting.

Expected Result:

Cyberbullying

Actual Result:

Alert generated

- Test Case 5

  Message:

  You are an idiot.

  Expected Result:

  Toxic Language

  Actual Result:

  Alert generated

  The system successfully detected harmful messages and displayed detailed AI-generated alerts in the Parent Dashboard.

Chapter – 9: Expenses

Total Cost: ₹0

Item	Quantity	Cost (₹)
Laptop	1	Existing Resource
Internet Connection	1	Existing Resource
Python Software	1	Free
Streamlit Framework	1	Free
JSON Database	1	Free
Visual Studio Code	1	Free

Chapter – 10: Conclusion

The **SafeConnect – AI Powered Child Safety Monitoring System** was successfully designed and developed to enhance children's online safety by monitoring digital conversations and detecting potentially harmful interactions. The system provides separate login portals for children and parents, ensuring secure communication while allowing parents to receive alerts about suspicious messages.

The project demonstrates how Artificial Intelligence can be used to identify cyberbullying, abusive language, and other online threats at an early stage. The implemented solution is simple, user-friendly, and capable of supporting parents in protecting their children without continuously monitoring every conversation manually.

Overall, the SafeConnect system successfully achieved its objectives by providing a secure child communication platform integrated with Artificial Intelligence for detecting cyberbullying, online grooming, and toxic language. The system offers separate login portals for parents and children, AI-based threat classification, severity analysis, threat summaries, and actionable recommendations. Future enhancements include multilingual support, cloud databases such as Firebase or MongoDB, real-time notifications, image and voice analysis, and deployment as a mobile application.

References

[1] Python Documentation – https://www.python.org

Used as the primary reference for learning Python programming concepts, syntax, libraries, and implementation of the project modules.

[2] Streamlit Documentation – https://streamlit.io

Referred to for designing the web-based user interface, creating dashboards, and implementing interactive features for parents and children.

[3] JSON Documentation – https://www.json.org

Used to understand JSON data structures for storing and retrieving chat messages and user information in the application.

[4] Research Articles on Cyberbullying Detection

Studied various research papers and journal articles to understand cyberbullying, online harassment, and existing methods for detecting harmful online conversations.

[5] NLP and Toxic Comment Detection Literature

Referred to research on Natural Language Processing (NLP) and AI-based toxicity detection techniques to understand how machine learning can identify abusive and suspicious messages.

[6] Hugging Face Transformers Library:

Used to access and integrate the Toxic-BERT model for AI-based detection of toxic and harmful chat messages.

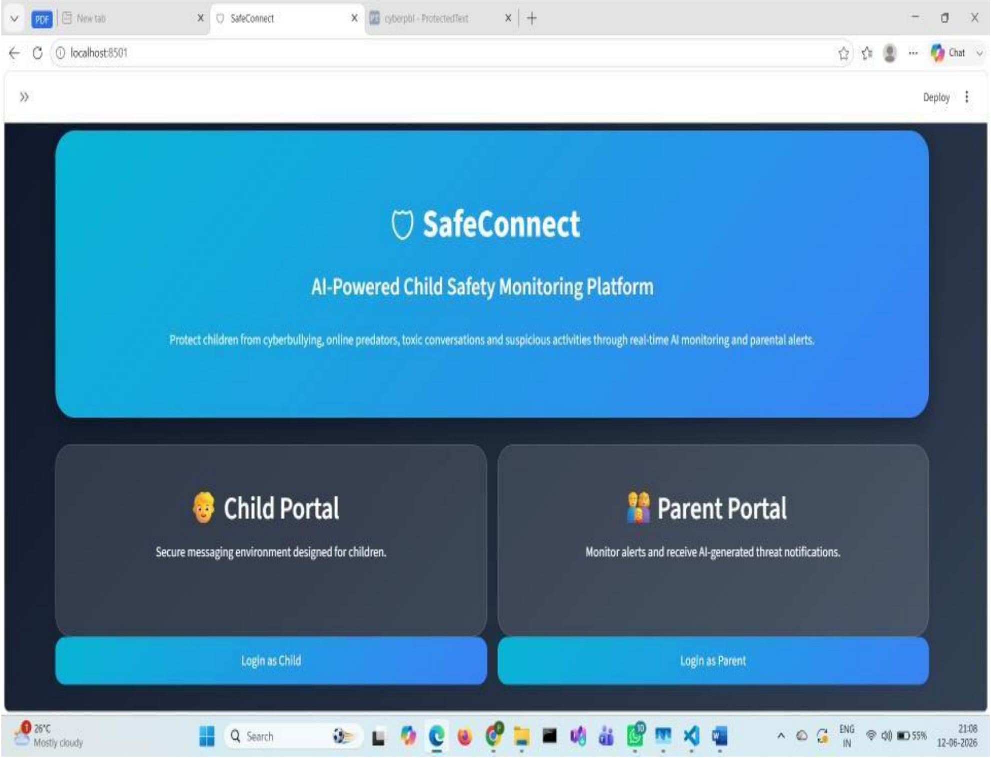
[7] Toxic-BERT (unitary/toxic-bert):

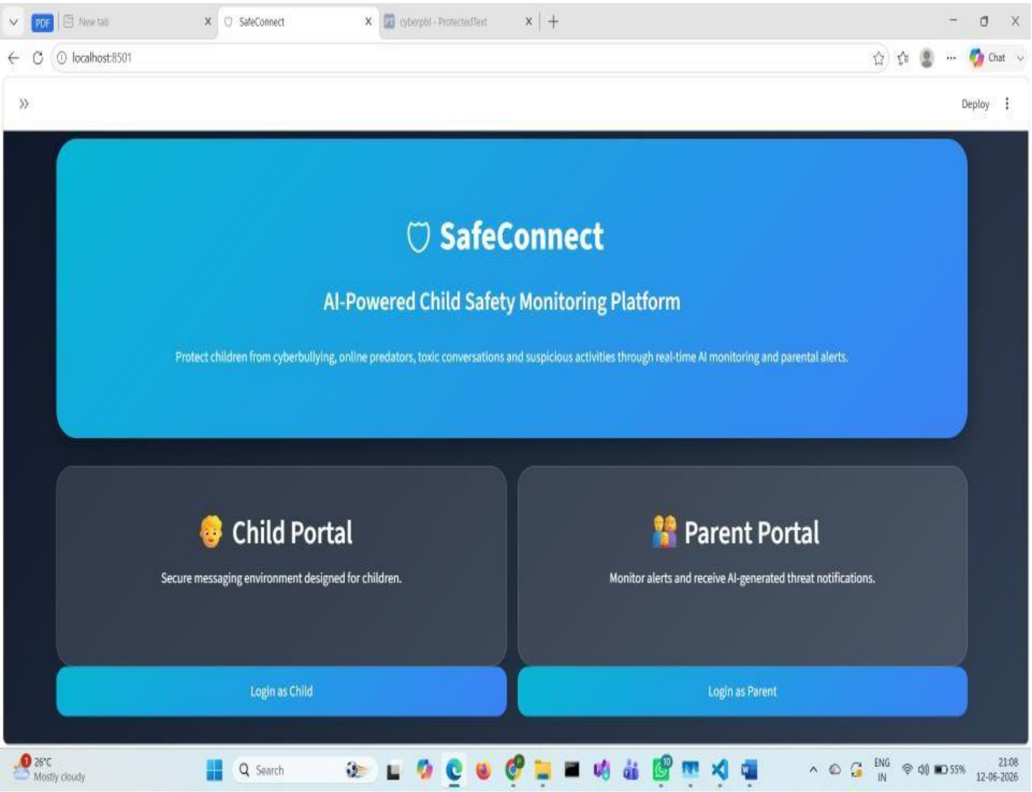
A pre-trained Natural Language Processing (NLP) model used to identify toxic, abusive, and offensive language in chat conversations

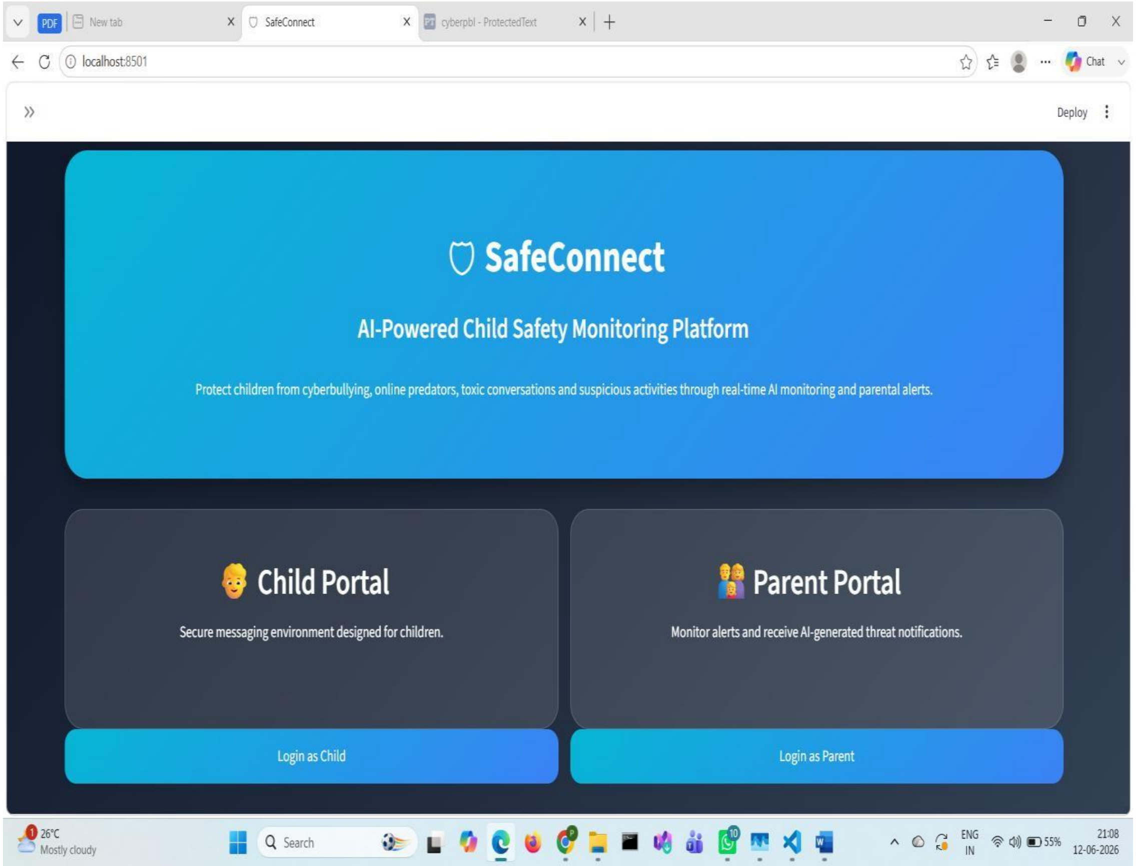
Appendix

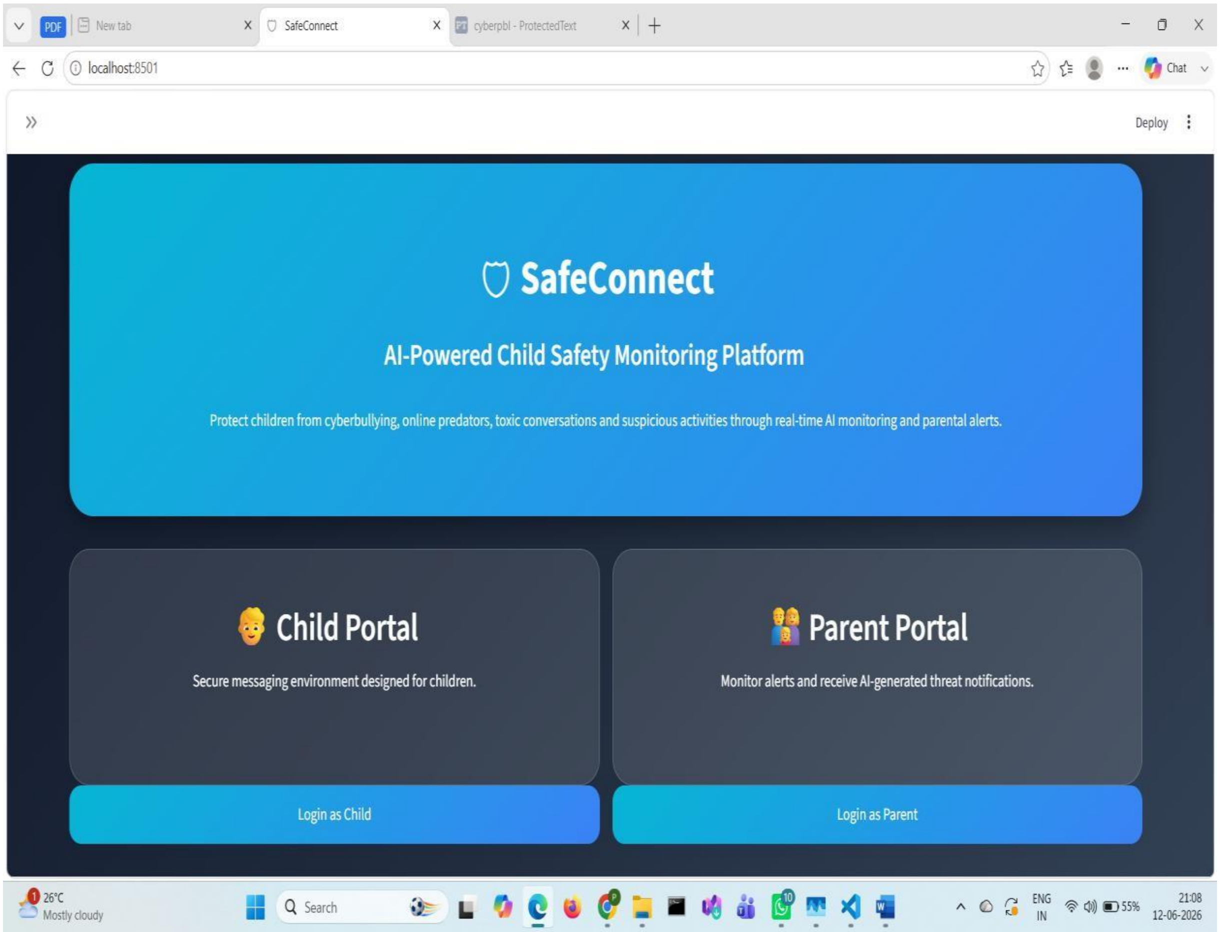
Appendices

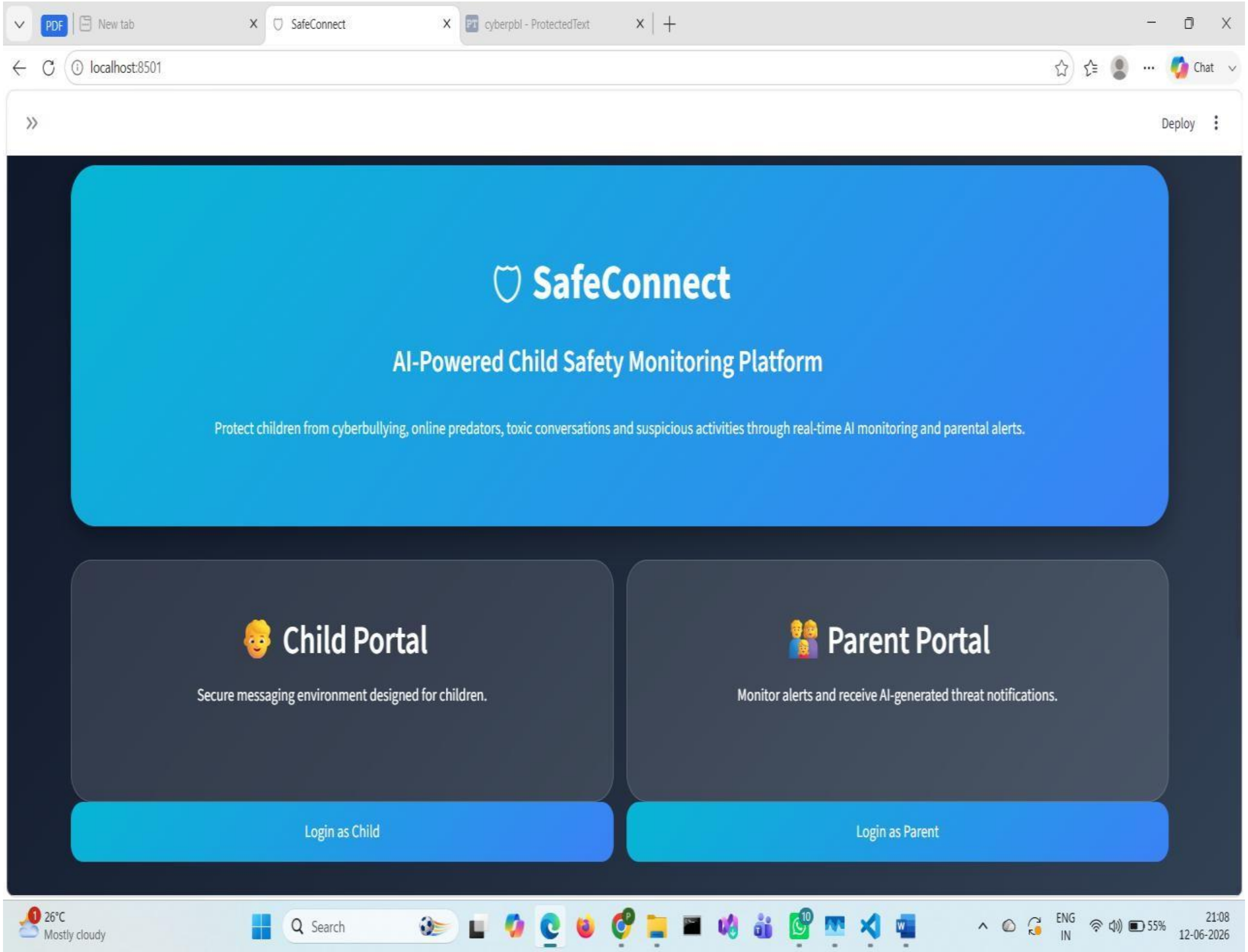
Appendix A: Login Screens



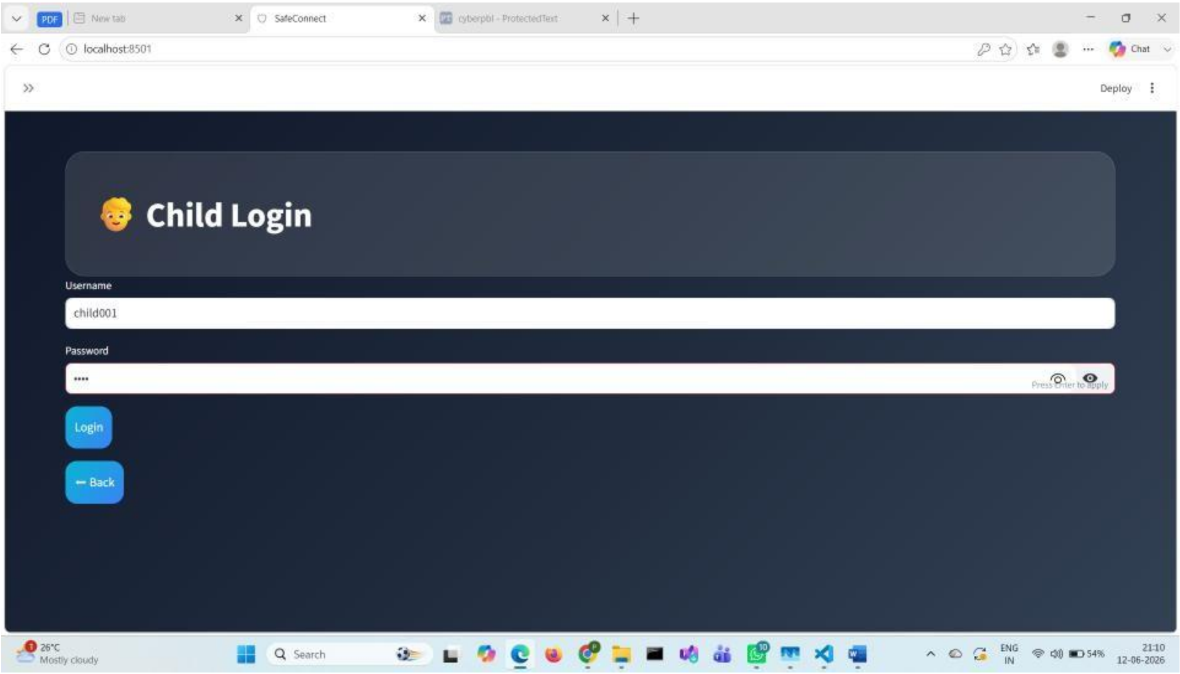


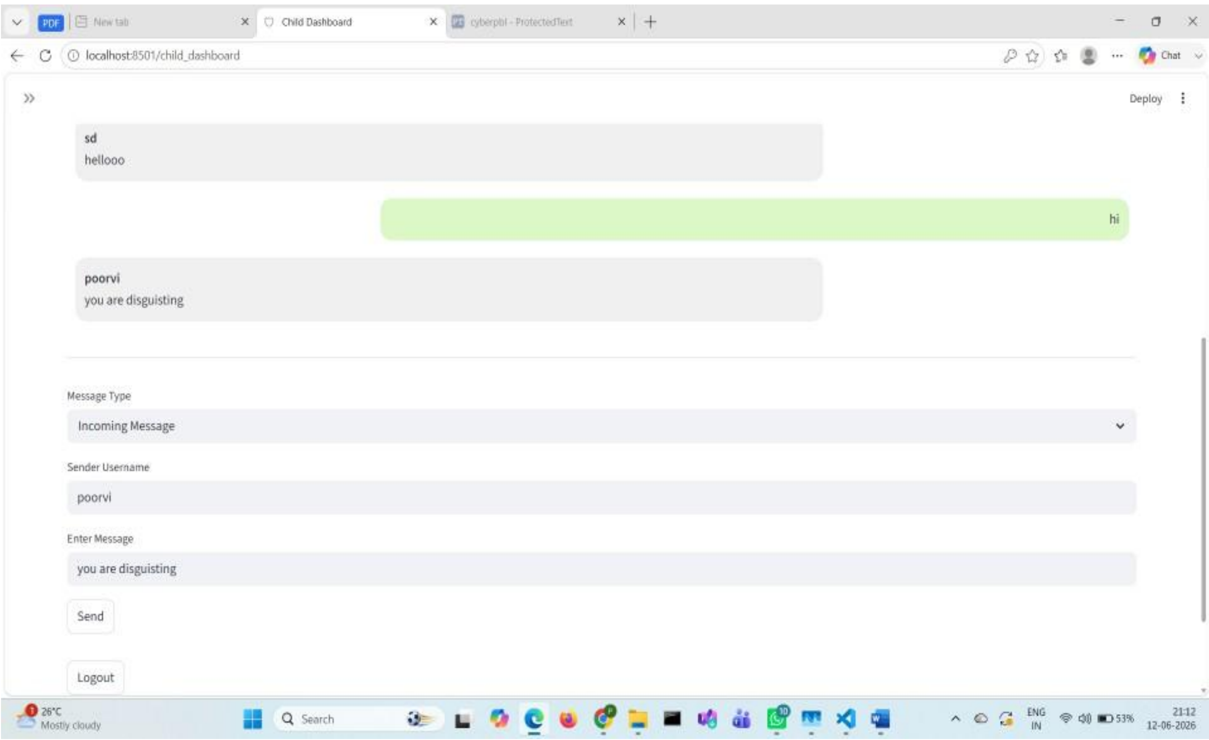

SafeConnect
AI-Powered Child Safety Monitoring Platform
Protect children from cyberbullying, online predators, toxic conversations and suspicious activities through real-time AI monitoring and parental alerts.
Child Portal
Secure messaging environment designed for children.
Parent Portal
Monitor alerts and receive AI-generated threat notifications.
Login as Child
Login as Parent


SafeConnect
AI-Powered Child Safety Monitoring Platform
Protect children from cyberbullying, online predators, toxic conversations and suspicious activities through real-time AI monitoring and parental alerts.
Child Portal
Secure messaging environment designed for children.
Parent Portal
Monitor alerts and receive AI-generated threat notifications.
Login as Child
Login as Parent

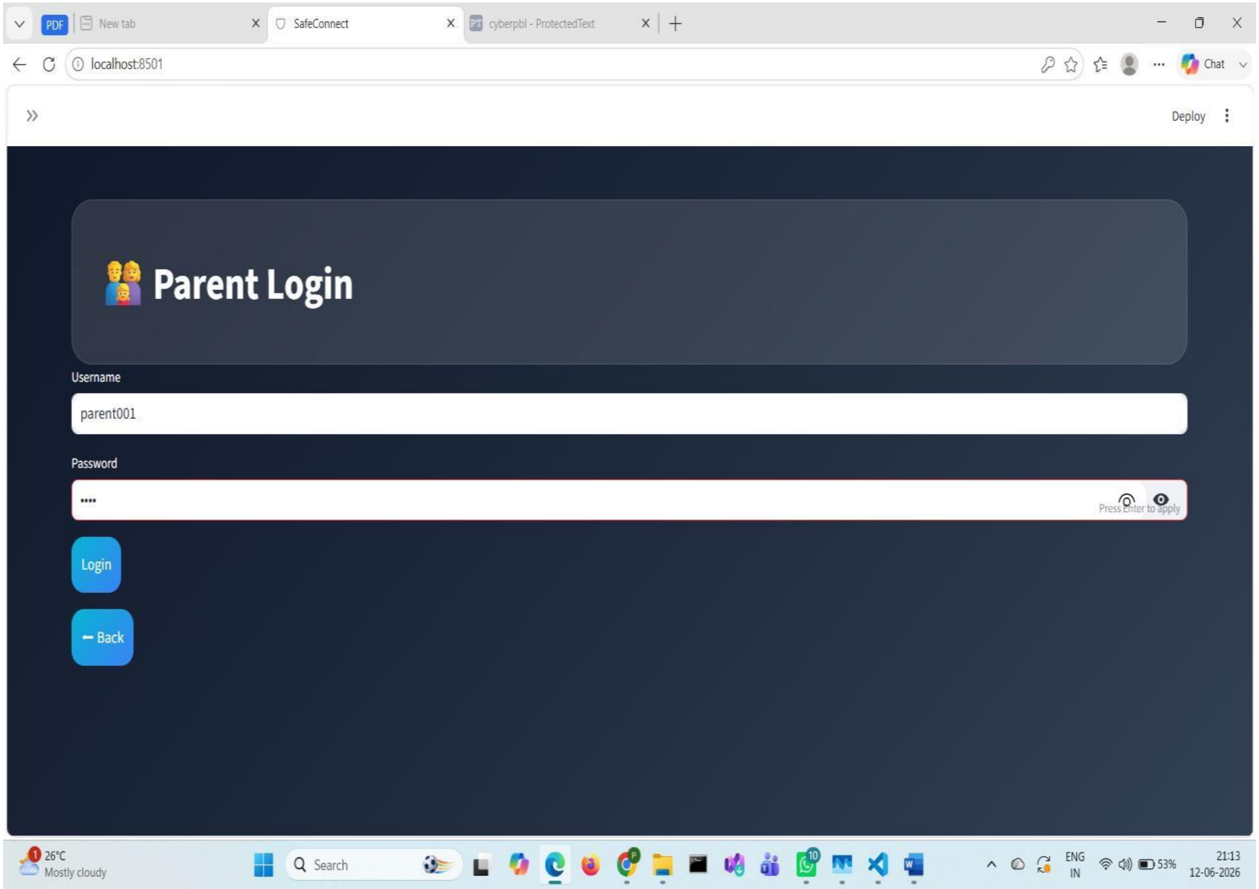

localhost:8501
Deploy
SafeConnect
AI-Powered Child Safety Monitoring Platform
Protect children from cyberbullying, online predators, toxic conversations and suspicious activities through real-time AI monitoring and parental alerts.
Child Portal
Secure messaging environment designed for children.
Parent Portal
Monitor alerts and receive AI-generated threat notifications.
Login as Child
Login as Parent

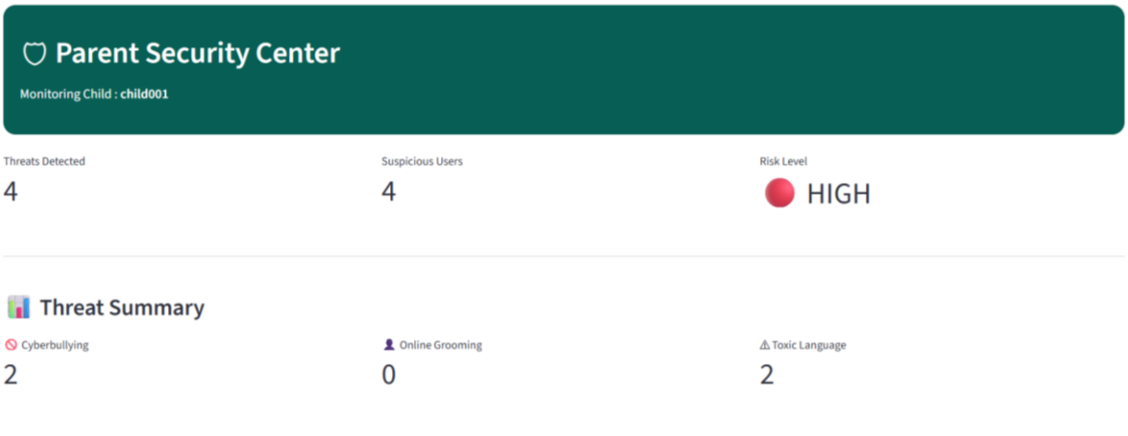
Appendix B: Child Dashboard Screenshot

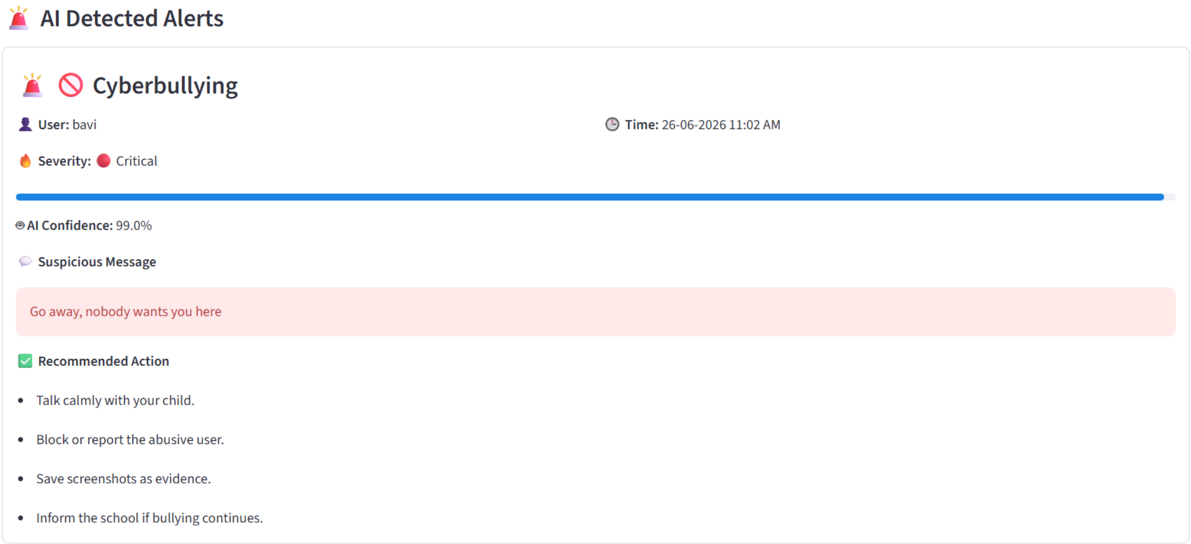


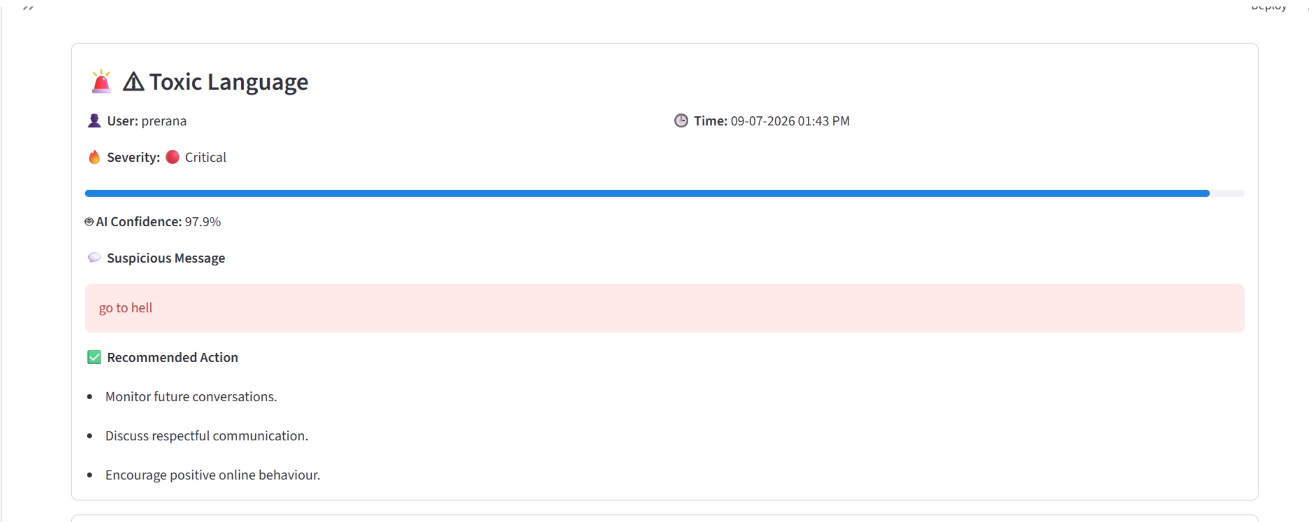


Appendix C: Parent Dashboard Screenshots






AI Detected Alerts
Cyberbullying
User: bavi
Time: 26-06-2026 11:02 AM
Severity: Critical
AI Confidence: 99.0%
Suspicious Message
Go away, nobody wants you here
Recommended Action
Talk calmly with your child.
Block or report the abusive user.
Save screenshots as evidence.
Inform the school if bullying continues.


Toxic Language
User: prerana
Time: 09-07-2026 01:43 PM
Severity: Critical
AI Confidence: 97.9%
Suspicious Message
go to hell
Recommended Action
Monitor future conversations.
Discuss respectful communication.
Encourage positive online behaviour.